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Investigation of fibre reinforced mud brick as a building material ☆

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Abstract

Most of the buildings in the rural areas are made out of limestone, low quality traditional concrete brick and adobe. But these materials do not have sufficiently high compressive strengths. In the present research, an earthquake-resistant material with high compressive strength has been sought. To this end, the mechanical properties of certain combinations of fibrous waste materials and some stabilisers were investigated thoroughly and some concrete conclusions were drawn. It was concluded that the interface layers of fibrous materials increased the compressive strength and a certain geometrical shape of these layer materials gave the best results. The mix proposed satisfies the minimum compressive strength requirements of ASTM and Turkish Standards.

Keywords: Fibre reinforced; Mud brick; Compressive strength; Straw; Polystyrene fabric

1. Introduction

Earth has been used in the construction of shelters for thousands of years and approximately 30% of the world's present population still lives in earthen structures [1]. Earth is a cheap, environmentally friendly and abundantly available building material. It has been used extensively for wall construction around the world, particularly in developing countries [2]. Home brick-makers have long been using fibrous ingredients like straw to improve the tensile strength of mud bricks. However, they have not had a chance to do scientific experimental investigation on the balance of ingredients and the optimisation of this production.

The fibres, which are connected together by mud, provide a tensile strength in mud bricks. The fibres provide a better coherence between the mud layers. The stress–strain relation of mud bricks under compression is very important. The compressive strength of fibre reinforced mud brick has been found to be higher than that of the conventional fibreless mud brick. Because, fibres are strong against stresses. In the mud brick, there are fibres in both the longitudinal and transverse directions. These fibres prevent the deformations that may appear in the mud brick, thus, preserving the shape of the brick, and preventing the regions near the surface from being crushed and falling off. Where there are fibres in the mud, the transverse expansion due to the Poisson's effect is prevented by the fibres. The existence of these fibres increases the elasticity of the mud brick. When the mud brick starts to dry, it deforms and contraction (shrinkage) takes place. The distribution of the fibres being arbitrary, as their number increases,

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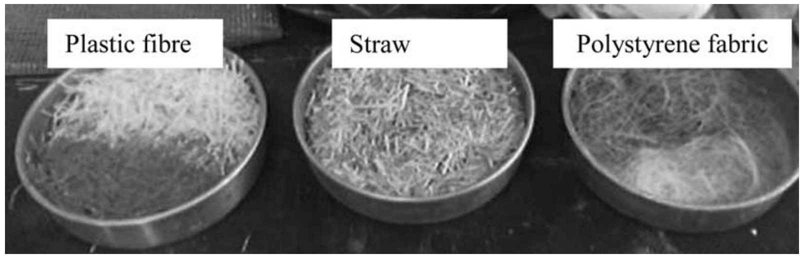

Fig. 1. Plastic fibre, straw and polystyrene fabric used in this study.

the tensile strength and elasticity properties of the mud brick improve. Thus, the mud brick behaves more flexible.

Most studies reported in the literature are focused on the stabilisation and utilisation of laterite and lateritic soils with the addition of lime, cement, or bentonite [3].

Southern Turkey is rich in natural pozzolans, which are also called 'trass' in the cement industry. Almost 155,000 km^2 of the country is covered by Tertiary and Quaternary-age volcanic rocks, among which tuffs occupy important volumes. Although there are many geological investigations on these volcanic rocks (pumice), their potential as natural pozzolans is not well established [4].

Housing is a great problem in today's world. In Turkey, many houses in rural areas are built with one floor. The most common building material for construction of houses is the usual burnt clay brick. Continuous removal of topsoil, in producing conventional bricks, creates environmental problems. In Cukurova region of Turkey, a huge quantity of straw is produced every summer. This is often a cause of major concern because farmers burn this material and give rise to ecological problems. Instead of burning, this material can be used in mud brick production. Similarly, plastic fibres and polystyrene fabric of vast amounts are produced in textile and plastic industries deteriorating the environment. Those materials will also serve as auxiliary materials in the production of fibre reinforced mud bricks.

Fibre reinforced mud brick design practice has been concerned with providing ductility to members deforming inelastically. Ductility capacity, is important only in its relation to ductility demand, and this can be expressed equivalently in terms of displacement capacity and demand.

Table 1
Chemical compositions of the cement, basaltic pumice, lime and gypsum (%)

	Cement	Basaltic pumice	Lime	Gypsum
SiO_2	20.1	43.9	–	–
Al_2O_3	5.2	14.1	–	–
Fe_2O_3	3.9	12.1	–	–
CaO	64.1	9.3	83.4	–
MgO	2.2	8.9	3.6	–
$Na_2O + K_2O$	1.4	0.3	–	–
SO_3	1.2	–	0.8	42.8
LOI	0.5	0.5	–	–
Crystal water	–	–	–	19.4
SiO_2 (non-solition)	–	–	0.5	–
$Al_2O_3 + Fe_2O_3$	–	–	0.5	–
CO_2	–	–	1.8	–

Table 2
Chemical composition and physical properties of clay [6]

Mineral			Concentration (meq/l)								Density (g/cm^3)	Permeability
			Cation					Anion				
Clay	Sand	Silt	Ca^+	Mg^+	Na^+	K^+	HCO_3^-	CO_3^-	CL^-	SO_4^-		
32.04	43.44	24.52	3.5	3.7	0.31	0.31	4.28	0.44	1.3	4.49	1.278	Quickly

Chemical composition

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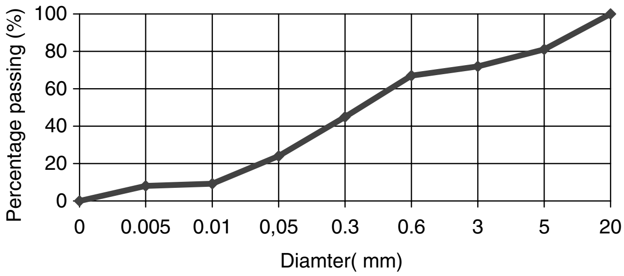


Fig. 2. Grading curve s of clay-basaltic pumice-cement used for the preparation of specimens (ASTM and Turkish Standard requirements).

This study elaborates on the compressive strength of fibre reinforced mud bricks made out of clay, cement, basaltic pumice, lime and gypsum using plastic fibre, straw, polystyrene fabric as fibrous ingredients, each at a time. The behaviour of the fibers in mud bricks and the effects of the different geometrical shapes of the interface layers were investigated in detail.

2. Experimental program

2.1. Materials

The materials used in this study for fibre reinforced mud brick production were clay as main matrix; cement, basaltic pumice, lime and gypsum as stabilisers; straw, plastic fibres and polystyrene fabric as fibrous materials and water as lubricant.

The basaltic pumice cone deposits are of Quaternary age and are located in the Cukurova region (Southern Turkey), and there are reserves estimated to be approximately 1000 million tonnes. The pumice comprises an average of 85% volcanic glass and 15% phenocrystic feldspars along with minor spheroid hematite minerals, determined by microscopy. XRD shows the presence of dominant illite and kaolinite as clay minerals along with feldspar. The high porosity of the basaltic pumice is an advantage for easy and economical crushing [5]. Plastic fibres, straw and polystyrene fabric used in this study are given in Fig. 1.

2.2. Chemical composition of the matrix

Chemical compositions of the cement, lime, gypsum and basaltic pumice used in this study are given in Table 1 and chemical composition and physical properties of clay are given in Table 2.

2.3. Process operation

2.3.1. Mixing of raw materials

The particle size analysis of the basaltic pumice and clay was made and the corresponding grading cure was obtained (see Fig. 2). The other materials mentioned in Table 3 were added to the mixture with the

Table 3
Designations of mixtures

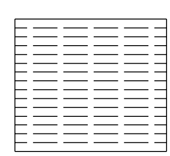
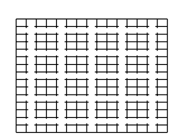
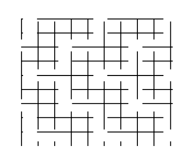
Mix designation	Composition of bodies of fibrous mud bricks	Geometrical shape of specimens		
		 1	 2	 3
A	Clay + pumice + cement + lime + gypsum + plastic fibre + water	A1	A2	A3
B	Clay + pumice + cement + lime + gypsum + straw + water	B1	B2	B3
C	Clay + pumice + cement + lime + gypsum + polystyrene fabric + water	C1	C2	C3
Traditional mud brick	Clay + straw + water	No geometrical shape	No geometrical shape	No geometrical shape

Table 4
Mixture proportions

Mix designations	Components (kg)								
	Clay	Cement	Basaltic pumice	Lime	Gypsum	Plastic fibre	Straw	Polystyrene fabric	Water
A	50	10	15	2	3	0.1	–	–	20
B	50	10	15	2	3	–	2	–	20
C	50	10	15	2	3	–	–	0.5	20

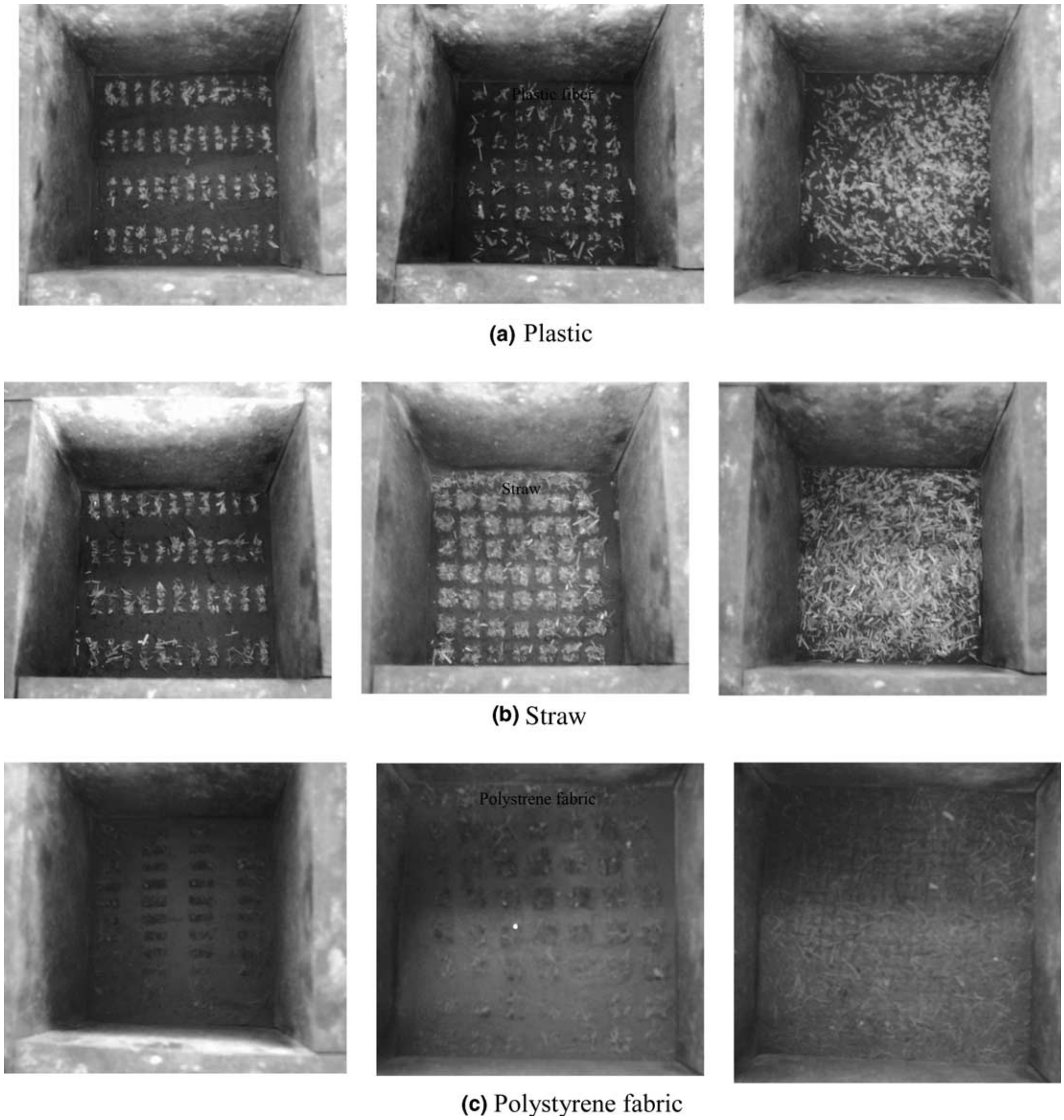


Fig. 3. Thin horizontal layer of plastic fibre (a), straw (b) and polystyrene fabric (c) placed at 1/3 and 2/3 heights.

proportions given in Table 4, to obtain three different specimen groups, and mixed thoroughly in dry state. Water was added and the ingredients were further mixed thoroughly by kneading until the mass attained a uniform consistency.

2.3.2. Preparation of bricks

The size of the fibre reinforced mud bricks to be used in the compression strength tests was chosen according to the Turkish standard, namely 150 mm × 150 mm × 150 mm cubes. The mixture was

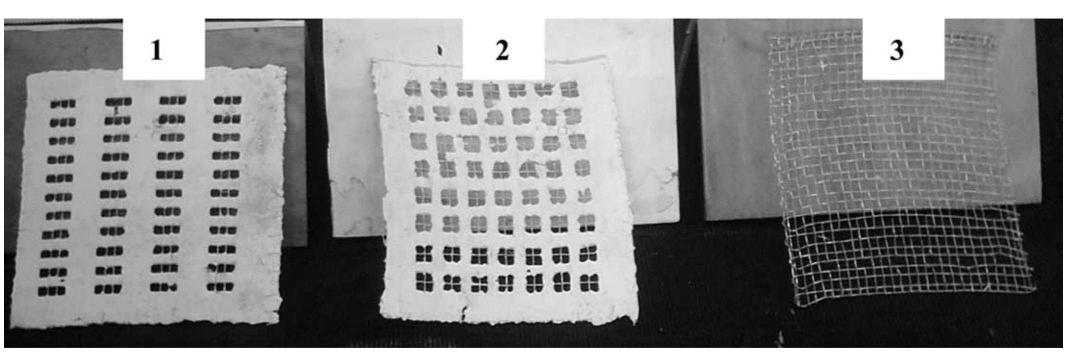


Fig. 4. The sievelike special gadget.

placed in three layers in steel moulds. A thin horizontal layer of plastic fibres, straw and polystyrene fabric was placed at 1/3 and 2/3 heights (see Fig. 3), using sievelike special gadgets seen in Fig. 4. After being filled in the foregoing manner, the moulds were properly compacted on a vibration table. Different sizes of fibre reinforced mud bricks and traditional concrete brick are given in Fig. 5.

2.3.3. Testing of mud bricks

The mud bricks were taken out from the moulds, covered with wet gunny bags and allowed to cure for a week. Then they were tested for compressive strength after 28, 72 and 96 days of casting. The test results were found as the average of the compressive strengths of five specimens. If the individual variation was more than ±5% of the average, the value was not considered in calculating the average value.

3. Results and discussion

The compressive strength and other mechanical properties of fibre reinforced mud bricks are given in Table 5. The compressive strength values required by the Turkish standard for traditional mud brick is 0.5–1 N/mm^2. However, the values in the case of the fibre reinforced mud bricks tested in the present study are much higher, namely 3.7–7.1 N/mm^2. In practical terms, this means using fibre reinforced mud bricks the thickness of the outer load bearing walls can be reduced substantially.

3.1. The effect of fibrous geometrical shapes on the compressive strength

The highest compressive strength at all ages has been found plastic fibers. Considering early age average compressive strength, the compressive strength of group A

Fig. 5. Different sizes of fibre reinforced mud bricks and traditional concrete brick.

Table 5
The compressive strength and physical properties of fibre reinforced mud bricks

Mix designations	N/mm^2			Water absorption after 24 h (%)	Loss of weight after 7 days (%)
	7 d	28 d	96 d		
A1	2.3	5.0	6.1	37.6	16.1
A2	2.4	4.9	6.5		
A3	2.7	5.6	7.1		
Average	2.4	5.1	6.5		
B1	1.6	3.8	5.0	36.8	14.2
B2	1.7	4.1	5.3		
B3	2.0	4.8	5.8		
Average	1.7	4.2	5.3		
C1	1.7	2.4	3.7	33.5	13.4
C2	2.1	2.4	4.2		
C3	2.0	2.6	4.9		
Average	1.9	2.4	4.2		
Traditional mud brick			2	38.7	17.5

specimens were found to be 17% higher than that of group B specimens and 21% higher than that of group C. Considering 96 days' average compressive strength, the compressive strength of group A specimens were found to be 19% higher than that of group B specimens and 36% higher than that of group C (see Table 5).

Specimens, which absorbed much water had higher density and weight lost than the others.

The effects of geometric shapes on the compressive strengths of different groups were different. Specimens in which geometric shape 3 was used, yielded the heights compressive strengths. Their compressive strengths were about 15% higher than those of group 2 specimens with geometric shape 1 and those of group 2 nearly mid way between the two (see Table 5). The compressive strengths of groups A and C mud bricks were, respectively, about two and three times that of traditional mud bricks.

3.2. The resistance to earthquakes

Most of the buildings in the rural areas of Turkey are made out of limestone, low quality traditional concrete brick (see Fig. 5) and adobe. But walls made of these materials do not have sufficient resistance to earthquakes. The compressive strength of the fibre reinforced mud bricks is greater than the traditional bricks, thus, being more resistant to earthquakes. Moreover, the presence of fibres in mud bricks provides flexibility to the structures thus enhancing their earthquake resistance.

Another one of the earthquake studies is the application of two-dimensional loading on a mud brick specimen. From the tests carried out, it was observed that, mud brick was strong enough, ductile and resistant against earthquakes [7]. Due to its fibers, fibre reinforced mud brick can store more elastic energy compared to other mud brick types, which renders it more resistant to earthquakes. For the same reason, fibre reinforced mud brick is more advantageous compared to the conventional mud brick.

4. Conclusions

Based on the experimental investigation reported in the paper the following conclusions can be drawn:

- Fibre reinforced mud bricks fulfill compressive strength requirements of ASTM and Turkish Standards.
- Group A specimens have higher compressive strength than the others.
- It was observed that the interface layers of fibrous materials increased the compressive strength and geometrical shape 3 gave the best result.
- Fibre reinforced mud bricks shall reduce the dead weight and materials handling cost for housing due to its comparatively higher compressive strength.
- This kind of mud brick can be moulded into any shape and size depending upon requirements, rendering it efficient as a building material.

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